

The Commercial Actors

SARS coronavirus is a new topic for many individuals. Since 1999, the ability to manipulate and exploit coronavirus for a variety of purposes has attracted the attention of individuals, institutions and commercial organizations in public, private, and not-for-profit sectors. The following is the list of over 5,100 patents and patent applications filed for the express purpose of controlling some aspect of the SARS coronavirus.

PATENT	Title	Owner	Priori ty	File d Dat e	Issu e Dat e
US9995706	Amperometric gas sensor	Steris Corporation	25- Jun- 12	30- Sep- 14	12- Jun- 18
US9995705	Amperometric gas sensor	Steris Corporation	25- Jun- 12	30- Sep- 14	12- Jun- 18
US9994558	Multicyclic compounds and methods of using same	Karyopharm Therapeutics Inc.	20- Sep- 13	19- Sep- 14	12- Jun- 18
US9994550	Heterocyclic modulators of lipid synthesis for use against cancer and viral infections	3-V Biosciences, Inc.	7- Jan- 14	7- Jan- 15	12- Jun- 18
US9993543	Immunogenic compositions comprising silicified virus and methods of use	Portland State University	31- Jan- 13	31- Jan- 14	12- Jun- 18
US9982257	Chiral control	WAVE LIFE SCIENCES LTD.	13- Jul-12	12- Jul-13	29- May-18
US9982241	Recombinant HCMV and RHCMV vectors and uses thereof	Oregon Health & Science University	14- May-10	1- Oct-15	29- May-18
US9982025	Monomeric griffithsin tandemers	The United States of America, as represented by the Secretary, Department of Health and Human Services	5- Jun-13	5- Jun-14	29- May-18
US9981036	Compositions, comprising improved IL-12 genetic constructs and vaccines, immunotherapeutics and methods of using the same	THE TRUSTEES OF THE UNIVERSITY OF PENNSYLVANIA	12- Dec-11	26- Feb-16	29- May-18
US9975885	Broad-spectrum non-covalent coronavirus protease inhibitors	PURDUE RESEARCH FOUNDATION	28- Apr-16	28- Apr-17	22- May-18
US9974850	Immunogenic compositions and uses thereof	BOARD OF REGENTS, THE UNIVERSITY OF TEXAS SYSTEM	25- Mar-15	25- Mar-16	22- May-18
US9974848	Tetanus toxoid and CCL3 improve DC vaccines	Duke University	14- Nov-13	14- Nov-14	22- May-18
US9974845	Combination of vaccination and inhibition of the PD-1 pathway	CureVac AG	22- Feb-13	21- Feb-14	22- May-18
US9970061	Bioagent detection oligonucleotides	IBIS BIOSCIENCES, INC.	27- Dec-11	27- Dec-12	15- May-18
US9969793	Compositions and methods for the treatment of immunodeficiency	ADMA Biologics, Inc.	28- Oct-14	13- Nov-17	15- May-18
US9963718	LCMV-GP-VSV-pseudotyped vectors and tumor-infiltrating virus-producing cells for the therapy of tumors	VIRATHERAPEUTICS GMBH	8- Oct-08	7- Apr-17	8- May-18
US9963611	Composition for use in decreasing the transmission of human pathogens	Innonix Technologies, Incorporated	29- May-09	21- May-10	8- May-18
US9963427	Dithiol mucolytic agents	PARION SCIENCES, INC.	23- Aug-13	11- Mar-16	8- May-18
US9962439	Injectable vaccine composition	NITTO DENKO CORPORATION	3- Oct-13	2- Oct-14	8- May-18
US9957302	Treating cancer with viral nucleic acid	Mayo Foundation for Medical Education and Research	20- Feb-07	6- Jul-15	1- May-18
US9957300	Virus-like particles, methods of preparation, and immunogenic compositions	Emory University	17- May-02	4- May-15	1- May-18
US9957238	Arylalkyl-and aryloxyalkyl-substituted epithelial sodium channel blocking compounds	Parion Sciences, Inc.	13- Dec-13	1- Mar-17	1- May-18
US9951317	Highly efficient influenza matrix (M1) proteins	NOVAVAX, INC.	11- Jul-03	6- Oct-16	24- Apr-18

US9951124	Antibody producing non-human mammals	MERUS N.V.	27-Jun-08	25-Jan-13	24-Apr-18
US9951122	Antibodies against influenza virus and methods of use thereof	BURNHAM INSTITUTE FOR MEDICAL RESEARCH	6-Dec-07	12-Aug-13	24-Apr-18
US9950062	Compounds and compositions as TLR activity modulators	GLAXOSMITHKLINE BIOLOGICALS SA	2-Sep-09	1-Sep-10	24-Apr-18
US9945856	Coronavirus, nucleic acid, protein, and methods for the generation of vaccine, medicaments and diagnostics	AMSTERDAM INSTITUTE OF VIRAL GENOMICS B.V.	18-Aug-03	13-Aug-14	17-Apr-18
US9945780	Use of a fluorescent material to detect failure or deteriorated performance of a fluorometer	GEN-PROBE INCORPORATED	14-Jun-12	7-Jun-13	17-Apr-18
US9944928	Construction of pool of interfering nucleic acids covering entire RNA target sequence and related compositions	York Yuan Yuan Zhu	23-Jul-07	2-Jul-15	17-Apr-18
US9944695	Antibody producing non-human mammals	Merus N.V.	27-Jun-08	30-Apr-14	17-Apr-18
US9944686	Treatment of tumors with recombinant interferon alpha	SUPERLAB FAR EAST LIMITED	28-Feb-01	5-Sep-13	17-Apr-18
US9944649	Compounds and compositions as toll-like receptor 7 agonists	Novartis Ag	1-May-14	29-Apr-15	17-Apr-18
US9943614	Cationic steroid antimicrobial diagnostic, detection, screening and imaging methods	BRIGHAM YOUNG UNIVERSITY	17-Jun-08	16-Jun-09	17-Apr-18
US9938300	Isothiazolopyrimidinones, pyrazolopyrimidinones, and pyrrolopyrimidinones as ubiquitin-specific protease 7 inhibitors	Forma Therapeutics, Inc.	5-Feb-15	4-Feb-16	10-Apr-18
US9938275	Substituted imidazoquinolines, imidazopyridines, and imidazonaphthyridines	3M Innovative Properties Company	18-Jun-04	23-Jan-17	10-Apr-18
US9938258	Substituted 2,3-dihydrobenzofuranyl compounds and uses thereof	Karyopharm Therapeutics Inc.	29-Nov-12	27-Nov-13	10-Apr-18
US9932351	Thienopyrimidinones as ubiquitin-specific protease 7 inhibitors	Forma Therapeutics, Inc.	5-Feb-15	4-Feb-16	3-Apr-18
US9932323	Therapeutic hydroxypyrimidinones, hydroxypyrimidinones and hydroxypyridazinones	Rutgers, The State University of New Jersey	11-Sep-12	13-Jan-17	3-Apr-18
US9931316	Antiviral activity from medicinal mushrooms and their active constituents	Not Available	31-Mar-15	14-Sep-15	3-Apr-18
US9926340	NAD analogs and methods of using said NAD analogs in determining ribosylation of proteins with PARP mutants	Biolog Life Science Institute Forshungslabor und Biochemica-Vertrieb GmbH	8-Apr-15	1-Apr-16	27-Mar-18
US9925215	Anionically modified polyallylamine derivative, use of anionically modified polyallylamine derivative as medicine, particularly for prophylaxis and treatment of infections of respiratory tract caused by human metapneumovirus (hMPV), human rhinoviruses (HRV), and infection by influenza virus type A (IAV) and pharmaceutical composition comprising the anionically modified polyallylamine derivative	UNIWERSYTET JAGIELLONSKI	29-Jul-14	25-Oct-17	27-Mar-18
US9920314	Compositions for and methods of identifying antigens	President and Fellows of Harvard College	21-Feb-06	6-May-15	20-Mar-18
US9920128	Synthetic antiserum for rapid-turnaround therapies	The Johns Hopkins University	28-Jan-15	20-Jan-16	20-Mar-18
US9919034	Methods of treating and prophylactically protecting mammalian patients infected by viruses classified in Baltimore group V	TAMIR BIOTECHNOLOGY, INC.	28-Mar-14	10-Jun-15	20-Mar-18
US9915613	Systems and methods for distinguishing optical signals of different modulation frequencies in an optical signal detector	GEN-PROBE INCORPORATED	24-Feb-11	21-Mar-14	13-Mar-18
US9914976	Methods and compositions for prostate cancer metastasis	FLORIDA AGRICULTURAL AND MECHANICAL UNIVERSITY (FA)	25-Mar-11	27-May-16	13-Mar-18
US9913801	Treatment of evolving bacterial resistance diseases including Klebsiella pneumoniae with liposomally formulated glutathione	YOUR ENERGY SYSTEMS, LLC	15-Feb-13	15-Mar-13	13-Mar-18
US9909176	Efficient deep sequencing and rapid genomic speciation of RNA viruses (vRNAseq)	The Johns Hopkins University	8-Sep-14	1-Sep-15	6-Mar-18
US9908946	Generation of binding molecules	Merus N.V.	26-Sep-11	16-Sep-15	6-Mar-18
US9908675	Powdered pouch and method of making same	MONOSOL, LLC	16-Apr-12	19-Jul-16	6-Mar-18

US9907796	Methods of treating tumoral diseases, or bacterial or viral infections	INHIBIKASE THERAPEUTICS, INC.	4- Oct- 12	15- Sep- 16	6- Mar- 18
US9895692	Sample-to-answer microfluidic cartridge	Micronics, Inc.	29- Jan- 10	5- Aug- 15	20- Feb- 18
US9895411	Analogues of C5a and methods of using same	BOARD OF REGENTS OF THE UNIVERSITY OF NEBRASKA	29- Jun- 10	29- Jun- 11	20- Feb- 18
US9895341	Inflammation and immunity treatments	Ocean Spray Cranberries, Inc.	1- Apr- 11	30- Mar- 12	20- Feb- 18
US9894888	Transgenic immunodeficient mouse expressing human SIRP-alpha	INSTITUT PASTEUR	26- Mar- 12	26- Mar- 13	20- Feb- 18
US9890419	Nanoreporters and methods of manufacturing and use thereof	NanoString Technologies, Inc.	23- Dec- 05	20- May- 16	13- Feb- 18
US9890408	Multiple displacement amplification	IBIS BIOSCIENCES, INC.	15- Oct- 09	15- Oct- 10	13- Feb- 18
US9890362	Compositions, methods and uses for inducing viral growth	Takeda Vaccines, Inc.	5- Dec- 08	19- Sep- 14	13- Feb- 18
US9890361	Methods for increasing the infectivity of viruses utilizing alkyne-modified fatty acids	LIFE TECHNOLOGIES CORPORATION	26- Jan- 12	25- Jan- 13	13- Feb- 18
US9890206	H1N1 flu virus neutralizing antibodies	Medigen Biotechnology Corporation	20- Aug- 15	20- Aug- 15	13- Feb- 18
US9890169	Triazolinone compounds as HNE inhibitors	CHIESI FARMACEUTICI S.P.A.	14- Dec- 15	12- Dec- 16	13- Feb- 18
US9890124	Benzazepine sulfonamide compounds	Hoffmann-La Roche Inc.	15- Dec- 15	14- Jun- 17	13- Feb- 18
US9889194	Immunogenic composition for MERS coronavirus infection	New York Blood Center, Inc.	1- Mar- 13	28- Feb- 14	13- Feb- 18
US9885092	Materials and methods for detection of HPV nucleic acids	QIAGEN GAITHERSBURG INC.	24- Feb- 11	23- Feb- 12	6- Feb- 18
US9885082	Embodiments of a probe and method for targeting nucleic acids	University of Idaho	19- Jul-11	19- Jul-12	6- Feb- 18
US9885037	Chiral control	WAVE LIFE SCIENCES LTD.	13- Jul-12	12- Jul-13	6- Feb- 18
US9884895	Methods and compositions for chimeric coronavirus spike proteins	The University of North Carolina at Chapel Hill	20- Mar- 14	20- Mar- 15	6- Feb- 18
US9884876	Anti-viral compounds, pharmaceutical compositions, and methods of use thereof	Kineta, Inc.	9- May- 14	8- May- 15	6- Feb- 18
US9884129	Release of agents from cells	The Brigham and Women's Hospital, Inc.	15- Oct- 09	5- Jan- 15	6- Feb- 18
US9884032	Esters of short chains fatty acids for use in the treatment of immunogenic disorders	PROPONENT BIOTECH GMBH	3- Oct- 12	3- Mar- 16	6- Feb- 18
US9884026	Modular particles for immunotherapy	YALE UNIVERSITY	1- Nov- 13	31- Oct- 14	6- Feb- 18
US9880151	Method of determining, identifying or isolating cell-penetrating peptides	Phylogica Limited	23- May- 11	23- May- 12	30- Jan- 18
US9879026	Substituted spirocycles	Boehringer Ingelheim International GmbH	12- Sep- 14	29- Nov- 16	30- Jan- 18
US9879003	Host targeted inhibitors of dengue virus and other viruses	Dana-Farber Cancer Institute, Inc.	11- Apr- 12	15- Mar- 13	30- Jan- 18
US9878988	Dendrimer like amino amides possessing sodium channel blocker activity for the treatment of dry eye and other mucosal diseases	PARION SCIENCES, INC.	29- May- 12	5- Jan- 16	30- Jan- 18
US9873678	Chemical compounds	AstraZeneca AB	18- Mar- 14	17- Mar- 15	23- Jan- 18
US9873674	C-Rel inhibitors and uses thereof	CORNELL UNIVERSITY	21- Sep- 12	19- Sep- 13	23- Jan- 18
US9872900	Nucleic acid vaccines	ModernaTX, Inc.	23- Apr- 14	5- Apr- 16	23- Jan- 18
US9872898	Compositions and methods for treating and preventing porcine reproductive and respiratory syndrome	Ohio State Innovation Foundation	24- Apr- 12	3- Oct- 16	23- Jan- 18
US9872895	TLR5 ligands, therapeutic methods, and compositions related thereto	Emory University	24- Sep- 10	20- Sep- 11	23- Jan- 18

US9868952	Compositions and methods for resistance-proof SiRNA therapeutics for influenza	Sirnaomics, Inc.	8-Jul-12	7-Jul-13	16-Jan-18
US9868740	Pyrimidinone compounds which are HNE inhibitors	CHIESI FARMACEUTICI S.p.A.	12-Jun-14	12-Jun-14	16-Jan-18
US9868736	Deubiquitinase inhibitors and methods for use of the same	THE REGENTS OF THE UNIVERSITY OF MICHIGAN	10-Oct-13	10-Oct-14	16-Jan-18
US9867882	Carbohydrate conjugates as delivery agents for oligonucleotides	Alnylam Pharmaceuticals, Inc.	4-Dec-07	25-Aug-15	16-Jan-18
US9867877	Methods for preparing squalene	NOVARTIS AG	12-May-10	22-Nov-16	16-Jan-18
US9862706	Compounds	CHIESI FARMACEUTICI S.p.A.	31-May-16	26-May-17	9-Jan-18
US9861614	Nuclear transport modulators and uses thereof	Karyopharm Therapeutics Inc.	9-May-12	23-Jun-15	9-Jan-18
US9856254	Alkoxy substituted imidazoquinolines	3M Innovative Properties Company	3-Oct-03	13-Jun-16	2-Jan-18
US9856241	Substituted benzofuranyl and benzoxazolyl compounds and uses thereof	Karyopharm Therapeutics Inc.	3-Jul-13	3-Jul-14	2-Jan-18
US9856228	Peptidyl nitril compounds as dipeptidyl peptidase I inhibitors	PROZYMEX A/S	9-Sep-13	8-Sep-14	2-Jan-18
US9856224	Stable sodium channel blockers	PARION SCIENCES, INC.	30-Jun-14	30-Jan-17	2-Jan-18
US9855287	Anti-viral azide containing compounds	LIFE TECHNOLOGIES CORPORATION	28-Jul-10	20-Aug-15	2-Jan-18
US9855284	Pharmaceutical compositions and methods	Pop Test Oncology LLC	3-Aug-15	6-Dec-16	2-Jan-18
US9849143	Broad spectrum antiviral and methods of use	The Burlington HC Research Group, Inc.	17-Apr-06	16-Feb-17	26-Dec-17
US9845342	Fusion proteins, recombinant bacteria, and methods for using recombinant bacteria	Spogen Biotech Inc.	17-Sep-14	17-Sep-15	19-Dec-17
US9840731	Preservation of biological materials in non-aqueous fluid media	Gentegra, LLC	14-Mar-13	14-Mar-14	12-Dec-17
US9840719	Variant AAV and compositions, methods and uses for gene transfer to cells, organs and tissues	The Children's Hospital of Philadelphia	22-Jul-13	22-Jul-14	12-Dec-17
US9840491	Quinazolinones and azaquinazolinones as ubiquitin-specific protease 7 inhibitors	FORMA Therapeutics, Inc.	5-Feb-15	4-Feb-16	12-Dec-17
US9839687	Acetylenedicarboxyl linkers and their uses in specific conjugation of a cell-binding molecule	SUZHOU M-CONJ BIOTECH CO., LTD.	15-Jul-15	15-Jul-15	12-Dec-17
US9834812	Probe kit for detecting a single strand target nucleotide sequence	Fondazione Istituto Italiano Di Tecnologia	27-Dec-12	27-Dec-13	5-Dec-17
US9834791	CRISPR-related methods and compositions with governing gRNAs	Editas Medicine, Inc.	7-Nov-13	7-Nov-14	5-Dec-17
US9834757	Hand, foot, and mouth vaccines and methods of manufacture and use thereof	Takeda Vaccines, Inc.	7-Nov-14	6-Nov-15	5-Dec-17
US9834595	Amino acid sequences directed against envelope proteins of a virus and polypeptides comprising the same for the treatment of viral diseases	Ablynx N.V.	5-Jun-08	29-Oct-15	5-Dec-17
US9833504	Virus-like particles and process for preparing same	Folia Biotech Inc.	13-May-11	1-May-12	5-Dec-17
US9833492	Combinations of a caspase inhibitor and an antiviral agent	Centre National de la Recherche Scientifique	2-Nov-07	15-May-15	5-Dec-17
US9832998	Antiviral compositions	Long Island University	30-May-07	19-Mar-15	5-Dec-17
US9828382	Pyrimidinone compounds as human neutrophil elastase inhibitors	Chiesi Farmaceutici S.p.A.	18-Dec-12	10-May-16	28-Nov-17
US9828379	Pyrrolo-pyrrole carbamate and related organic compounds, pharmaceutical compositions, and medical uses thereof	ABIDE THERAPEUTICS, INC.	3-Jul-13	1-Jul-14	28-Nov-17
US9828370	Compositions and methods for inhibiting kinases	INHIBIKASE THERAPEUTICS, INC.	23-Apr-15	22-Apr-16	28-Nov-17
US9828346	N-myristoyl transferase inhibitors	University of Dundee	2-Sep-08	31-Aug-15	28-Nov-17

US9828342	Isatin derivatives, pharmaceutical compositions thereof, and methods of use thereof	CITY OF HOPE	24-Feb-12	25-Feb-13	28-Nov-17
US9827190	Intradermal delivery of immunological compositions comprising toll-like receptor 7 agonists	GLAXOSMITHKLINE BIOLOGICALS SA	1-Feb-13	30-Jan-14	28-Nov-17
US9822339	Means and methods for influencing the stability of antibody producing cells	ACADEMISCH MEDISCH CENTRUM BIJ DE UNIVERSITEIT VAN AMSTERDAM	9-Dec-05	26-Aug-15	21-Nov-17
US9822173	Heterodimeric immunoglobulins	AMGEN INC.	21-Nov-12	21-Nov-13	21-Nov-17
US9822165	Hydrocarbon stapled stabilized alpha-helices of the HIV-1 GP41 membrane proximal external region	DANA-FARBER CANCER INSTITUTE, INC.	18-Jun-09	18-Jun-10	21-Nov-17
US9822155	Method of preventively treating a subject at the risk of developing infections of a respiratory virus	Xiangxue Group (Hong Kong) Company Limited	9-May-13	23-Aug-16	21-Nov-17
US9822127	GAK modulators as antivirals	The Board of Trustees of the Leland Stanford Junior University	23-Jul-14	23-Jul-15	21-Nov-17
US9822065	Benzazepine dicarboxamide compounds	Hoffmann-La Roche Inc.	6-Mar-15	14-Feb-17	21-Nov-17
US9821052	Reverse genetics systems	Seqirus UK Limited	31-Jul-09	30-Jul-10	21-Nov-17
US9821051	Reducing hospitalization in elderly influenza vaccine recipients	Seqirus UK Limited	28-Oct-10	21-Oct-11	21-Nov-17
US9816078	Compositions for increasing polypeptide stability and activity, and related methods	SOLIS BIODYNE OÃœ	19-Nov-09	11-Mar-16	14-Nov-17
US9815886	Compositions and methods for the treatment of immunodeficiency	ADMA BIOLOGICS, INC.	28-Oct-14	8-Jan-15	14-Nov-17
US9815805	Certain (2S)-N-[(1S)-1-cyano-2-phenylethyl]-1,4-oxazepane-2-carboxamides as dipeptidyl peptidase 1 inhibitors	ASTRAZENECA AB	24-Jan-14	8-Nov-16	14-Nov-17
US9814777	Targeting lipids	Arbutus Biopharma Corporation	4-Dec-07	22-Oct-13	14-Nov-17
US9810683	Use of live cell interferometry with reflective floor of observation chamber to determine changes in mass of mammalian cells	The Regents of the University of California	6-May-09	25-Nov-13	7-Nov-17
US9809845	Methods and reagents for amplifying nucleic acids	The United States of America, as represented by the Secretary, Department of Health and Human Services	6-Aug-12	6-Aug-12	7-Nov-17
US9809796	Animal protein-free media for cultivation of cells	Baxalta GmbH	29-Oct-04	18-May-17	7-Nov-17
US9809632	Universal protein tag for double stranded nucleic acid delivery	University of Washington Through its Center for Commercialization	23-Oct-13	22-Oct-14	7-Nov-17
US9809591	Heterocyclic modulators of lipid synthesis	3-V Biosciences, Inc.	8-Mar-11	5-Oct-15	7-Nov-17
US9808490	Induced hepatocytes and uses thereof	ACCELERATED BIOSCIENCES CORP.	26-Nov-14	25-Nov-15	7-Nov-17
US9803236	Microarray-based assay integrated with particles for analyzing molecular interactions	CapitalBio Corporation	6-Aug-10	6-Aug-10	31-Oct-17
US9803197	Particle-nucleic acid conjugates and therapeutic uses related thereto	Emory University	25-Jun-12	27-Feb-13	31-Oct-17
US9802937	Substituted pyrazolo{4,3-D}pyrimidines as kinase inhibitors	ORIGENIS GMBH	21-Apr-11	23-Apr-12	31-Oct-17
US9802919	Compounds	CHIESI FARMACEUTICI S.p.A.	31-May-16	26-May-17	31-Oct-17
US9801948	Antimicrobial compositions and methods of use thereof	Yale University	21-Sep-11	21-Sep-12	31-Oct-17
US9801947	Methods and compositions for enhancing immune response	3M INNOVATIVE PROPERTIES COMPANY	10-Apr-03	6-Oct-14	31-Oct-17
US9801935	Soluble needle arrays for delivery of influenza vaccines	SEQIRUS UK LIMITED	20-Aug-10	11-Oct-16	31-Oct-17
US9801897	Delivery of RNA to trigger multiple immune pathways	GLAXOSMITHKLINE BIOLOGICALS SA	6-Jul-10	6-Jul-11	31-Oct-17
US9797000	Non-target amplification method for detection of RNA splice-forms in a sample	QIAGEN GAITHERSBURG INC.	1-May-09	30-Apr-10	24-Oct-17
US9796979	Oligonucleotide modulators of the toll-like receptor pathway	Quark Pharmaceuticals Inc.	3-Mar-11	28-Jul-16	24-Oct-17